DATA STRUCTURES - INTERVIEW REVISION SHEET

1. Array vs Vector

Array: No bounds check, fixed size, unsafe.
Vector: Dynamic, safe (.at()), heap allocation.

2. Stack vs Heap

Feature	Stack	Heap
Size	Small	Large
Allocation	Automatic	Dynamic
Risk	High (overflow)	Low

3. Map vs Unordered Map

Feature	map	unordered_map
DS	Red-Black Tree	Hash Table
TC	O(log n)	O(1) avg
Order	Sorted	Not sorted

4. Hash Table

Key → Hash → Bucket → Value.
Collision handled by chaining.
Avg O(1), Worst O(n).

5. Set vs Unordered Set

Feature	set	unordered_set
Order	Sorted	Not sorted
TC	O(log n)	O(1) avg
Duplicates	No	No

6. Stack & Queue

Stack: LIFO → push, pop, top → O(1).
Queue: FIFO → push, pop, front, back → O(1).

7. Deque

Double-ended queue → insert/delete from both ends in O(1).

8. Priority Queue

Uses Binary Heap.
Top: O(1), push/pop: O(log n).

9. Binary Heap

Complete binary tree + heap property.
Array representation.
Insert = heapify up, Delete = heapify down.

10. Multiset & Multimap

Multiset: Sorted + duplicates allowed.
Multimap: Multiple values per key.

■ Interview One-Liners

- map uses Red-Black Tree → O(log n) - unordered_map uses hashing → O(1) avg - Heap gives fastest min/max access - Complete tree ensures log n height - Stack/Queue operate in O(1)